ACID FM · SOUND · FX · MOD

engines, effects, modulation, sequencer and songs

firmware v0.4 · M-VAVE FM-1 · norfolkhacker.github.io/acid-fm

ENGINES · EDIT › SOUND / TONE

SEVEN ENGINES, ALL SYNTHESIS

ENGINE	SOUND (KIT)				TONE (SOUND)				WHAT TO KNOW
DRUM-X	MORPH	TUNE	TONE	DECAY	DRUM	LEVEL	PITCH	DECAY	8 sounds BD SD CP CH OH TM RS CB, sides A / B: **MORPH** moves the whole kit between them. KIT is the whole kit; SOUND (shown here) and SOUND 2 (DRUM NOISE COLOR WAVE) one drum, KNOB 1 picks it (a key struck too). MORE: SIDE, PITCH MOD, MOD TYPE, NOISE TYPE, CRACK, DRIVE, CLANG, FEEDBACK, SEND; the kit's NOISE, CLICK, FOLD, METAL, HANDS, STEREO, MORPH LFO.
VOICE	VOWEL	VOWEL 2	TALK	SHIFT	BUZZ	BREATH	RESO	RANDOM	A formant voice: vowels, talk, breath. **WORDS** (MORE) ES / EN / JA sings syllables, a note each: SOUND shows POSITION, THEME, CRISP; LYRIC is the text or the dice's phrases (TYPE LYRIC opens NAME; the web editor sends it too).
PHASE	WAVE	WAVE 2	BRIGHT	SWEEP	DETUNE	LINE 2	SUB	FEEDBACK	Phase distortion: BRIGHT how far the phase is bent, SWEEP how far the envelope moves it; FEEDBACK from hard edge to growl.
LOFI	CHIP	WAVE	DUTY	CRUSH	CUTOFF	VIBRATO	BEND	MASK	A chip: PLS TRI SAW NOIS WRAM. **BYTE**: 32 bytebeat FORMULAS, VARIANT · MOTION GRIT BEND LOOP (LOOP high: a tuned note). **FLOAT**: 32 floatbeat formulas in the key played; CHIP sets the resolution. RETRIG in MORE.
ANALOG	WAVE	DETUNE	MIX	NOISE	CUTOFF	RESO	DRIVE	INTERVAL	SAW SQR TRI SIN PWM **SYNC RING SAW3**; INTERVAL: osc 2 ±24 st. Its filter is per voice; page 3 is VOICE (VOICE GLIDE PRIORITY SPREAD).
FM6	ALGORITHM	FEEDBACK	MOD LEVEL	MOD RATIO	OP	RATIO	FINE	LEVEL	Six operators, 32 algorithms drawn (P 5 = the patch's own). TONE here is OPERATOR (ratio or FIXED freq, fine, level); OP ENV: OP STAGE RATE LEVEL. MORE: PATCH F1–F8, OWN, B1–B32 (a 32-voice .syx sent from the web editor), MOD TIME, MOD VEL, DETUNE.
NOISE	SOURCE	COLOR	FREQ	PEAK	KEY TRACK	DENSITY	DRIFT	CRUSH	Coloured and metallic noise (SOURCE ANLG DUST LFSR META), tracked to the key or free.

FILTER · EVERY ENGINE

FILTER	CUTOFF	RESO	DISTORT

LP · HP · BP · COMB. COMB rings tuned to the note: CUTOFF moves it, RESO how much it rings. Page 3 on PHASE, LOFI, VOICE, NOISE; on ANALOG, CUTOFF RESO DRIVE are TONE's and the type is in MORE; on FM6 and DRUM-X the filter heads MORE.

VOICE · PLAYING

VOICE	GLIDE	PRIORITY	SPREAD

POLY · MONO · LEGATO · UNISON; GLIDE and GLIDE MODE; LAST / LOW / HIGH priority. **12 voices** with both cores (8 with one), at most 8 a track.

DICE · GEN

EDIT + SELECT: a random factory preset of the engine, every setting moved up to a quarter of its range (DRUM-X: both sides of every sound around the factory kit). Heard at once: OCT+ keeps it, OCT– brings the sound back. SEQ / LFO / FX + SELECT: the sequence, the LFO, the effects, the same way. **HOME + SELECT**: GEN, HOME's last three pages: GENRE (EXPLORE, RANDOM, or IDM, BROKEN BEAT, BREAKS, DRILL'N'BASS, ACID, ELECTRO, MICRO, RETROBOC, AMBIENT, DRIFT) · SEED · ENERGY · FORM; DRUMS · BASS · CHORDS · ARP; LEAD · COMPLEX · SWING · HUMAN. OCT+ makes the piece, OCT– back to TRACKS. GLO + G4 undoes.

MODULATION

ENV · LFO

ENV › AMP

ATTACK	DECAY	SUSTAIN	RELEASE

ENV › ENV TO

FILTER	PITCH	SHAPE	LOOP

LOOP: held, the envelope cycles from the attack: an ADSR-shaped LFO.

LFO

RATE	WAVE	MODE	FADE

SINE TRI SAW SQR RAND · TRIG / FREE

LFO › DEST

PITCH	FILTER	SHAPE	AMP

MOD MATRIX · 4 SLOTS A TRACK

SLOT	SOURCE	DEST	AMOUNT

SOURCES

LFO · ENV · VELOCITY · KEY · NOTE RAND · WHEEL · PRESSURE · EXPR · MACRO 1–4 · STEPS · NOISE

DESTINATIONS

PITCH · FILTER · SHAPE · AMP · PAN · DISTORT · CHORUS · DELAY · REVERB · LFO RATE · VIBRATO · E1...E8 · CLIP · PUNCH · DLY TIME · DLY FDBK · DLY TONE · DLY MIX · REV SIZE · REV DAMP · CHO RATE · CHO DEPTH

E1...E8: the engine's own EDIT knobs, by their names (CUTOFF ...). CLIP / PUNCH: the master. DLY REV CHO: the buses.

STEPS · MACROS

LFO › MORE › STEPS · A 16-LEVEL SEQUENCER

LENGTH, STEP RATE, SMOOTH, STEP 1–16, in time with the transport. Use it from MOD MATRIX, SOURCE STEPS.

MACROS M1–M4

Hold LFO: the knobs are M1–M4, latched until LFO again (also HOME › VISUAL). F3 rolls new routes, G3 clears them. By hand: MOD MATRIX, SOURCE MACRO 1–4. Saved with the project.

SEQUENCER

64 steps a track, up to 4 notes a step. Four pages: **STEPS** (the roll; on DRUM the lane grid; KNOB 1 all 64 steps, SELECT on the last cell the next 16), **STEP EXTRAS** (the picked step's RATCHET, CHANCE, CONDITION EVERY 2ND / 1ST OF 4 ..., MICRO in 64ths), **PATTERN**, **MORE**. **Hold a step + turn a knob**: a lock (LOCK · SETTING · VALUE). Step recording: stopped, REC on STEPS; a chord held + KNOB 1 its length, + ARP arpeggiated. MORE › STEP: two components a step (CHANCE, RATCHET, ARP, SEQ, WALK, PULSE, OCTAVE, RANDOM, BEND, TONAL, JUMP, REVERSE, CONDITION, FILL, KILL) and EVERY. TRACK: SEQ PITCH, CHANCE thins the whole track, HUMAN, REC SNAP. SHAPE: STRETCH, REPEAT, NUDGE, ROTATE as you turn. REC + knobs while playing: automation (64 records a section; MORE › AUTOMATION plays, clears and lists them).

STEPS · STEP EXTRAS · PATTERN

STEP	NOTE	ACC/SLD	GATE
RATCHET	CHANCE	CONDITION	MICRO
LENGTH	DIVISION	SWING	GATE

SCALE · ARP

ROOT and SCALE are every melodic track's: CHROMATIC MAJOR MINOR DORIAN MIXOLYDIAN PENTATONIC MINOR PENT HARMONIC MINOR PHRYGIAN LYDIAN LOCRIAN MELODIC MINOR BLUES WHOLE TONE and two DIMINISHED. SNAP puts the keys (or the sequence) on the scale, **ALL TRACKS** moves every sequence along it. **CHORD**: a key plays a chord (TRIAD 7TH ADD9 SUS2 SUS4 in key, MAJ MIN 7 MAJ7 MIN7 POWER), VOICING LEAD leads each to the next, STRUM as a harp.

ARP MODES

OFF UP DOWN UP-DOWN RANDOM REPEAT DOWN-UP UP+OCTAVE CONVERGE DIVERGE PINKY THUMB WALK CHORD **TRNS** (the keys move the sequence)

SONGS

4 projects × 4 sections (A–D), a song of up to 16 rows by bars (SAVE › ARRANGE, OCT+ adds a row, PLAY plays; END: STOP, x2–x8, LOOP), SLOOP style. Each row can hold a **scene**: mutes, DRUM-X group mutes, macros and punch-ins. SAVE held plays sections live on the next bar, stores them, records the song (SONG REC). Each section keeps its sounds, steps, automation, kit and its SLICER.

EFFECTS

PER TRACK

FX › SENDS

DISTORT	CHORUS	DELAY	REVERB

MORE › DISTORT

DIST TYPE	DIST TONE	–	–

SOFT · HARD · FOLD · CRUSH · RECTIFY · AMP.

MORE › AUTO PAN

PAN DEPTH	PAN RATE	PAN SYNC	PAN WAVE

The track's pan swung L–R around its PAN. SYNC **BEAT**: in time with the bar; **FREE**: in Hz.

SLICER · ONE ON A BUS

SLICE MODE	SLICE PAT	SLICE RATE	SLICE DEPTH

OFF · GATE · STUTTER, 16 patterns. **SLICE T1–T6** and **SLICE FX** (the delay / reverb / chorus returns) ON / OFF: everything on = the whole mix chopped after the effects. **SLICE PITCH**: the repeats' 16-step pitch (PITCH 1–16, PITCH LOOP, PITCH SNAP to the scale). Not with LOAD ECO.

DELAY · CHORUS

FX › SPACE

DLY TIME	DLY FDBK	REV TYPE	REV SIZE

MORE › DELAY

DLY TONE, DLY MIX, DLY WIDE, DLY PITCH, DLY SPRAY, DLY MOD, DLY TYPE, DLY PING. Synced or free (up to ~3 s); PITCH / SPRAY: a grain delay.

MORE › CHORUS

CHO RATE	CHO DEPTH	CHO WIDE	–

Not with LOAD ECO.

REVERB · FIVE MODELS

REV TYPE	REV SIZE	REV DAMP	REV PRE
REV MOD	REV RATE	REV TONE	REV WIDE

TYPE	WHAT IT IS	SIZE	DAMP	MOD	RATE	WIDE
ROOM	Classic room	decay	darkens	chorus on the tail	its speed	stereo
SPRING	Amp spring tank	length · decay	darkens	wobble	wobble speed	stereo
SHIMMER	Up an octave on every pass; **REV AIR** (±24 st: down darker and lower, up airier and higher)	decay	darkens	how much shimmer	—	—
RESO	4 strings tuned to the scale; **REV PITCH** (0–14 steps up), **FOLLOW** in DAMP's place (OFF / T1–T6: the chord follows that track's notes)	how long they ring	FOLLOW	past half: the chord opens	—	—
CLOUD	Granular wash; SIZE max **freezes** it; **REV AIR** (±24 st: the cloud down / up)	decay · freeze	darkens	grains pitched + chorus	grain density	grain spread

MASTER

CLIP	PUNCH	DUCK	DUCK TIME

CLIP saturates without getting louder. **PUNCH** drives the drums: denser body. **DUCK**: the kick pulls the rest down (DUCK TIME its release). An always-on **leveler** before the limiter keeps patches at a steady volume. **DJ FILTER** (GLO › LIVE KNOB 4): left a low-pass, right a high-pass.

PUNCH-IN FX (FX HELD)

REPEAT 1/8 · 1/16 · 1/32, GLITCH, TAPE STOP, FREEZE, SHUFFLE; on the notes: OCT–/+, ½ TEMPO, DECAY –/+, STUTTER, ATTACK +, ARP, RANDOM; sweeps CRUSH THROW. Knobs: STEP LOOP, INTERVAL, MUTATE, DJ FILTER. Recordable per section (FX › MORE › PUNCH LANE).

GPL-3.0 · Built on JIANT FM (Juan Jiant) and Felucca (Hügelton Instruments) · song mode from SLOOP